Project Report

Stochastic Process [MA2213]

Mixed Copula Model for Insurance Claims and Claim Sizes

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1 Introduction & Background

1.1 Problem Statement: Modeling Total Insurance Loss

In non-life insurance, one of the fundamental challenges is the accurate estimation of **total loss**, which is essential for setting premiums, assessing reserves, and pricing reinsurance contracts. Traditionally, this is done by modeling two key components: the *number of claims* and the *average claim size*. However, a naive summation of these two independent processes often falls short in capturing the complexities of real-world insurance data, where dependencies may exist between the frequency and severity of claims.

1.2 The Classical Compound Poisson Model

A commonly used approach to model total insurance loss is the **compound Poisson model**, originating from the work of Lundberg (1903). In this model:

- The number of claims is modeled using a Poisson distribution.
- The claim sizes are modeled independently using a Gamma distribution.

A critical assumption here is the *independence* between the claim count and the claim size.

While analytically convenient, this independence assumption is often violated in practice. For example, policyholders who file more claims might also tend to file larger claims, introducing a **positive dependency** between frequency and severity.

1.3 Introducing Copulas for Dependency Modeling

To address this limitation, we turn to **copulas**, which allow us to model *dependency structures separately from marginal distributions*. Copulas provide a powerful and flexible tool to capture the joint behavior of random variables, regardless of their individual distributions. In particular, *Gaussian copulas* are often used due to their mathematical tractability and interpretability.

However, standard copula approaches may not be sufficient when dealing with *mixed data types*—for example, discrete data (like number of claims) and continuous data (like claim size). This is where **mixed copula models** come into play.

1.4 Goal of the Paper

The goal of the paper is to develop and validate a **mixed copula model** that links:

- A Poisson generalized linear model (GLM) for the number of claims,
- A Gamma GLM for the average claim size, and
- A Gaussian copula to capture the dependency between them.

This approach allows for a joint modeling of claim frequency and severity, enabling more accurate estimation of total loss and better decision-making in insurance pricing and risk management.

1.5 Why Mixed Copulas Are Needed

Mixed copula models are essential because they:

- Handle discrete-continuous variable combinations (claim count and size),
- Preserve marginal GLM structures, allowing interpretability and flexibility,
- Introduce dependency modeling without violating GLM assumptions, and
- Provide superior fit and predictive performance compared to classical models, as shown in simulation studies and real-world car insurance data analysis.

2 Mathematical Model

2.1 Claim Frequency Model – Poisson GLM

We model the number of insurance claims per policyholder using a Poisson regression:

$$Y_{i2} \sim \text{Poisson}(\mu_{i2})$$
$$\log(\mu_{i2}) = \log(e_i) + \mathbf{z}_i^\top \boldsymbol{\beta}$$

- Y_{i2}: Number of claims for policy i
- μ_{i2}: Expected number of claims
- e_i: Exposure (e.g., policy duration, driving distance)
- $\mathbf{z}_i$: Covariates (e.g., driver age, region, vehicle type)
- $\boldsymbol{\beta}$: Regression coefficients

This is a generalized linear model (GLM) using the log-link function and Poisson distribution.

2.2 Claim Severity Model – Gamma GLM

The average claim size is modeled using a Gamma GLM:

$$Y_{i1} \sim \text{Gamma}(\mu_{i1}, v)$$
$$\log(\mu_{i1}) = \mathbf{x}_i^\top \boldsymbol{\alpha}$$

- Y_{i1}: Average claim size for policy i
- μ_{i1}: Expected claim size
- v: Dispersion parameter

- $\mathbf{x}_i$: Covariates for severity
- $\boldsymbol{\alpha}$: Regression coefficients

Gamma distribution is suitable for modeling positive, skewed data like claim sizes.

2.3 Dependency Modeling – Gaussian Copula

To capture the dependence between frequency and severity, we use a Gaussian copula.

2.3.1 Copula Theory

Sklar's Theorem allows expressing a joint distribution as:

$$F(y_{i1}, y_{i2}) = C(G_1(y_{i1}), G_2(y_{i2}))$$

- G_1, G_2: Marginal CDFs of Gamma and Poisson
- C: Copula function capturing dependence

2.3.2 Gaussian Copula Definition

$$C(u, v; \rho) = \Phi_2(\Phi^{-1}(u), \Phi^{-1}(v); \rho)$$

- Φ^{-1}: Inverse of standard normal CDF
- Φ_2: Bivariate standard normal CDF with correlation ρ
- $u = G_1(y_{i1})$, $v = G_2(y_{i2})$
- ρ: Copula correlation parameter

2.3.3 Final Joint Model

$$F(y_{i1}, y_{i2}) = \Phi_2\left(\Phi^{-1}(G_1(y_{i1})), \Phi^{-1}(G_2(y_{i2})); \rho\right)$$

2.4 Summary

- Claim Frequency: Modeled via Poisson GLM
- Claim Severity: Modeled via Gamma GLM
- Dependence: Captured using Gaussian copula

This structure enables flexible modeling of discrete-continuous insurance data while capturing joint behavior realistically.

3 Parameter Estimation via Maximization by Parts

Estimation Method: Maximization by Parts (MBP) Algorithm

Overview of the MBP Algorithm

The **Maximization by Parts (MBP)** algorithm is a numerical optimization technique designed to estimate parameters in complex models where standard Maximum Likelihood Estimation (MLE) is computationally intensive or unstable. In the context of the mixed copula regression model, MBP provides an efficient and stable method for estimating the regression parameters of the marginal models, as well as the dependency parameter from the Gaussian copula.

Log-likelihood Decomposition

The joint model includes both marginal distributions (Poisson for claim counts and Gamma for average claim sizes) and a dependency structure modeled by a Gaussian copula. The conditional log-likelihood is decomposed into two parts:

- **Marginal Part** $l_m(\boldsymbol{u}_1)$: Involves the regression parameters $\boldsymbol{a}$ (for the Gamma GLM) and $\boldsymbol{b}$ (for the Poisson GLM).
- **Dependency Part** $l_d(\boldsymbol{u}_1, \gamma)$: Involves the copula correlation parameter ρ, which is transformed using Fisher's z-transformation.

$$\ell(\boldsymbol{u}) = l_m(\boldsymbol{u}_1) + l_d(\boldsymbol{u}_1, \gamma)$$

where $\boldsymbol{u}_1 = (\boldsymbol{a}, \boldsymbol{b})$ and $\gamma = \frac{1}{2}\ln\left(\frac{1+\rho}{1-\rho}\right)$.

Parameter Updates in MBP

The MBP algorithm iteratively updates parameters using two key steps:

1. **Update $\boldsymbol{a}$ and $\boldsymbol{b}$ (marginal parameters)**: These are updated using Fisher scoring:

$$\boldsymbol{u}_1^{(k)} = \boldsymbol{u}_1^{(k-1)} + \left[\mathcal{I}_m(\boldsymbol{u}_1^{(k-1)})\right]^{-1} \nabla l_m(\boldsymbol{u}_1^{(k-1)})$$

   where $\mathcal{I}_m$ is the Fisher information matrix for the marginal part, and ∇l_m is the gradient.

2. **Update γ (copula parameter)**: The correlation parameter ρ is estimated indirectly via Fisher's z-transformation:

$$\gamma = \frac{1}{2}\ln\left(\frac{1+\rho}{1-\rho}\right)$$

   The score function with respect to γ is solved using a one-dimensional root-finding method (e.g., bisection) to update ρ.

Why MBP Instead of Regular MLE?

The MBP algorithm is preferred over standard MLE in this setting for several reasons:

- **High dimensionality**: The model includes a large number of parameters (e.g., over 60 covariates), making second-order optimization methods computationally expensive.
- **Mixed data types**: The presence of discrete (Poisson) and continuous (Gamma) components complicates the joint likelihood.
- **Copula dependency**: Introducing a copula adds non-standard dependence that violates assumptions typically required for standard MLE convergence.
- **Efficiency**: MBP leverages the separability of the likelihood to simplify optimization and avoid full Hessian computation.
- **Stability**: Simulation studies show MBP performs well in terms of convergence and bias, especially in small to moderate sample sizes.

4 Simulation Study

Purpose

This study is designed to test the performance of the mixed copula model in controlled synthetic environments. Specifically, it aims to assess the **bias**, **accuracy**, and **stability** of parameter estimates under varying levels of dependence and marginal behavior.

Experimental Design

A total of **24 scenarios** were simulated, each differing in:

- Regression parameters (a_1, a_2) and (b_1, b_2)
- Copula correlation $\rho \in \{0.1, 0.5, 0.9\}$
- Gamma dispersion parameter $v \in \{0.5, 1.0\}$

Each scenario simulated a dataset of **1000 observations**. To ensure robustness, each was repeated **500 times**.

Statistical Model Components

- **Gamma GLM:** $Y_{i1} \sim \text{Gamma}(\mu_{i1}, v), \quad \log(\mu_{i1}) = \mathbf{x}_i^\top \boldsymbol{\alpha}$
- **Poisson GLM:** $Y_{i2} \sim \text{Poisson}(\mu_{i2}), \quad \log(\mu_{i2}) = \log(e_i) + \mathbf{z}_i^\top \boldsymbol{\beta}$
- **Gaussian Copula:** $F(y_{i1}, y_{i2}) = C(G_1(y_{i1}), G_2(y_{i2}) \mid \rho)$

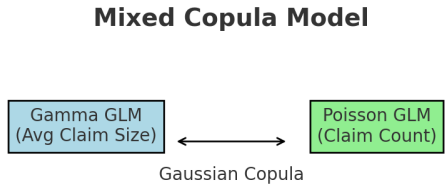

Figure 1: Structure of Copula-Based Regression Model

Results Summary

Table 1: Bias Comparison in Different Simulation Scenarios

Scenario	Max Bias (%) – Fixed ρ	Max Bias (%) – Adaptive ρ
Low ρ	1.28	1.21
Medium ρ	6.04	6.17
High ρ	14.08	15.74

5 Application to Real Data

Dataset Overview

A real dataset of **12,850 German car insurance policies** from the year 2000 is used. It contains:

- Average claim size (DM)
- Number of claims (minimum 1 per policy)
- Exposure time
- Covariates: Region, age, deductible, car age, premium rate, driving distance

Table 2: Number of Claims Distribution

Claims	Count	% Share
1	12472	97.1
2	356	2.77
3	20	0.16
4	2	0.02

Dependency Visualization

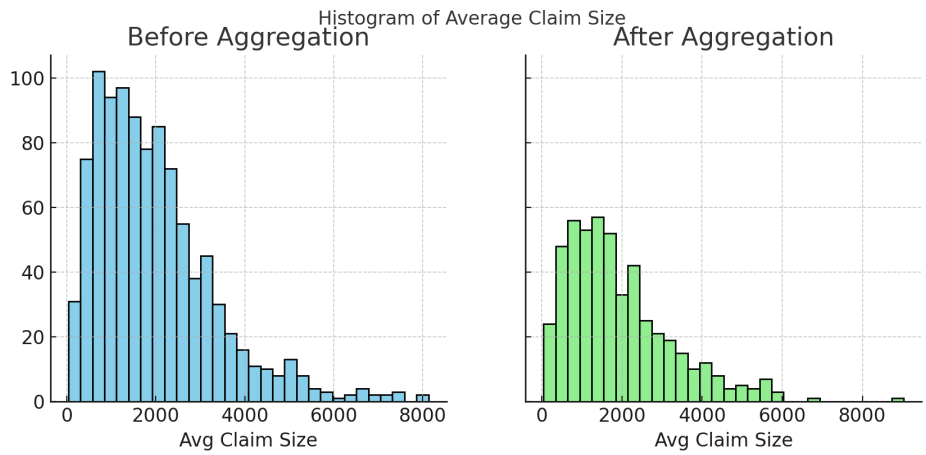

Figure 2: Histogram of Average Claim Size (Before vs After Aggregation)

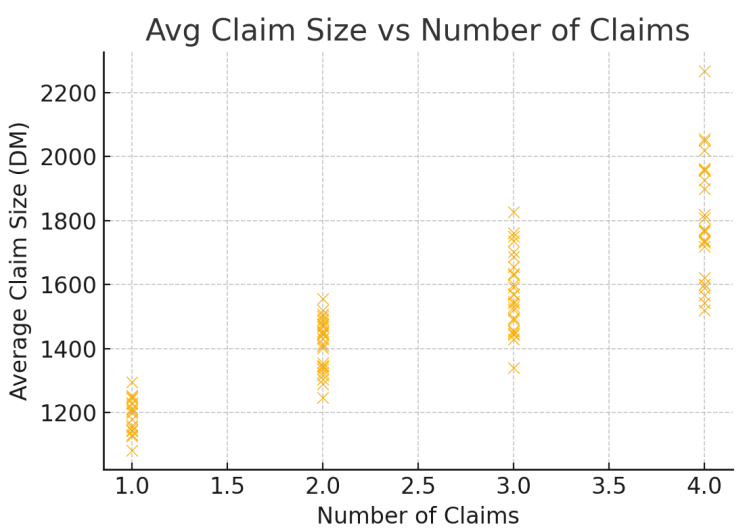

Figure 3: Avg Claim Size vs Number of Claims

Conclusion

The mixed copula model successfully identifies a small but meaningful dependency between claim frequency and severity. This violates the classical assumption of independence in compound Poisson models and improves risk evaluation and pricing decisions.

6 Conclusion

This project illustrates the power of mixed copula models in capturing dependencies between frequency and severity in insurance claims. The Gaussian copula, combined with GLMs and the MBP estimation algorithm, provides a flexible and robust framework.

Future work may explore:

- Heavy-tailed copulas (e.g., Student-t)
- Zero-inflated or hurdle models
- Real-time claim forecasting applications

7 references

https://www.cs.rice.edu/~vardi/papers/vmcai07.pdf